***Plants, People, Planet* Supporting Information**

Article title: Climate-driven intraspecific shifts in seed germination phenology: Consequences for native temperate woodland restoration and regeneration

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Methods S1 Details of germination and seed mass assessment

Germination experiments were conducted in two phases. In the first phase, which started in December 2021, one seed lot per species was tested (Table S1) at a range of constant temperatures (from 5 to 40°C, in 5°C increments), with and without the application of one month of cold stratification (28 days at 5°C in the dark). At the end of the stratification period, both cold-stratified and non-stratified seeds were placed, on the same day, into incubators (one per temperature; ± 2 °C precision) under a 12-h light photoperiod. Each condition included three to four replicates of 40 to 50 seeds, depending on availability, sown in 90mm Petri dishes with 1% agar-water solution. Germination (i.e., visible radicle protrusion) was recorded daily, excluding weekends, until the cumulative germination curve reached a plateau, after which germination was recorded twice per week and then weekly as germination rate slowed. The experiment was closed when no germination occurred at a given temperature for two consecutive weeks, after a minimum of four weeks and up to 63 days, when on average at least 50% of germination had occurred at the lowest temperature (5°C). A cut test was then performed to classify the remaining seeds as fresh but ungerminated, dead/mouldy, or embryoless. Embryoless seeds were excluded from further analysis due to their lack of potential to germinate, while all others were retained.

Based on the results achieved in this first phase, experiments of the second phase were then carried out on five more seed lots for each species, between April 2023 and February 2025. Prior to the germination experiment, seeds were weighed. For *A. glutinosa*, five replicates of 20 seeds were weighed at the population level. For the other two species, seed mass was measured at the mother tree level (see 'Study Species and seed collection' in the main text), using up to five replicates (depending on availability) of 20 seeds per tree for *B. pubescens*, and 20 individual seeds per tree for *P. sylvestris*. To reconstruct population-level representation across five replicates, a weighted average was calculated based on the proportion of seeds each tree contributed to the total number used in the experiment. The median seed weight was then calculated from these five replicates. For each species, germination trials for all populations and treatments started on the same day. In the germination experiment, *B. pubescens* was tested under the same conditions as in phase one while the temperature range was extended to 0°C for *A. glutinosa* and 45°C for *P. sylvestris*. Germination was recorded periodically at intervals designed to capture its progression across both treatments and various temperatures. For example, measurements were taken twice a day at temperatures near the optimum, while at other temperatures, recordings were made once daily or less frequently as needed. The same criteria as in phase one were used to determine the schedule and closing of the experiment, which lasted up to 277 days. A cut test was performed as in phase one.

Methods S2 Details of data analysis

All data processing and statistical analyses were conducted in R version 4.4.2 (R Core Team, 2024).

Germination percentage

To assess the effect of cold stratification on final germination percentage and potential germination failure under future thermal stress, we analysed total germination percentages at six weeks, during the period when all tests were ongoing. A separate quadratic logistic regression model (binomial GLM with logit link) was fitted for each combination of population and treatment to capture the unimodal thermal response and keep predictions within 0–100%. For each dataset, we recorded y_i seeds germinating out of n_i total germinable seeds sown at temperature T_i. We assume y_i follows a binomial distribution with parameters n_i and p_i, and that the logit of p_i is a quadratic function of temperature:

(Eqn. 1) $logit(p_i) = \beta_0 + \beta_1 T_i + \beta_2 T_i^2$

From the estimated coefficients β_0, β_1 and β_2, we derived key germination parameters analytically following the approach on Sentinella et al. (2020). The temperature of maximum predicted germination T_{pmax}, which is the vertex of the parabola, given by:

(Eqn. 2) $T_{pmax} = -\frac{\beta_1}{2\beta_2}$

Lower and upper bounds of T_{pmax} (T_{pmax_lower} and T_{pmax_upper}, respectively), defined as the temperature range at which the model predicts 95% of maximum predicted germination (p_{max}) on the logit scale:

(Eqn. 3) $T_{pmax_lower}, T_{pmax_upper} = -\frac{\beta_1 \pm \sqrt{\beta_1^2 - 4\beta_2[\beta_0 - logit(0.95 \times p_{max})]}}{2\beta_2}$

To investigate the future warming risk to seed-based regeneration, we calculated the difference between T_{pmax_upper} and the projected mean temperature of the warmest three months under the pessimistic scenario for each population (Sentinella et al., 2020). This allowed the assessment to reflect the highest plausible risk rather than conditions under average seasonal temperatures.

Germination rate and cardinal temperatures

To assess the germination thermal traits and to predict germination phenology of each population, we employed a thermal time modelling approach which is widely recognised for predicting

seed emergence in the field (Garcia-Huidobro et al., 1982; Covell et al., 1986; Ellis et al., 1986, 1987; Pritchard & Manger, 1990; Pritchard et al., 1999; Hardegree, 2006). In this framework, non-dormant seeds accumulate thermal units (e.g., °Cd) above a base temperature (T_b), below which the germination rate is zero. Accumulation proceeds until the optimal temperature (T_o), at which rate is maximal, and then declines until reaching a ceiling temperature (T_c), above which germination ceases. Here, we extend the model to seeds with non-deep PD (Carruggio et al., 2021) since the temperatures that break dormancy are considerably lower than those that enhance germination rate, thereby minimising the confounding effects of temperature on dormancy release versus germination.

Germination rates (defined as the reciprocal of the time taken for 50% of seeds to germinate) were estimated for each combination of replicate, temperature, treatment, population, and species. This was done using the *drmte* function and the nonparametric maximum likelihood estimator (*NPMLE*) for censored time-to-event data, followed by extraction of 50th percentile via the *quantile* function in the *drcte* R package (Onofri et al., 2022). These estimates were then fitted to temperature data (using the *drm* function from the *drc* R package (Ritz et al., 2015), with the self-starting *GRT.Ex* exponential model function (Catara et al., 2016; Masin et al., 2017) as implemented in *drcSeedGerm* R package (Onofri et al., 2018) to calculate thermal time (ϑ_T, °Cd) and cardinal temperatures for 50% germination (T_b: base, and T_c: ceiling temperature). The optimum temperature (T_o) was subsequently estimated via model predictions as the temperature where the rate was maximal. The exponential model was chosen over other parametrisations because it better reflected the typical pattern for these species, with rates rising steadily at sub-optimal temperatures and declining sharply at supra-optimal ones. Unlike the thermal time model in Garcia-Huidobro, Monteith & Squire (1982), the exponential model does not include separate thermal time parameters for sub- and supra-optimal intervals. However, the estimated ϑ_T, functions as an analogue to the sub-optimal ϑ_T, as it reflects thermal accumulation from T_b up to T_o (Catara et al., 2016; Masin et al., 2017).

Shifts in germination phenology under future climate scenarios

Since using monthly average temperatures could oversimplify thermal accumulation patterns, hourly temperature fluctuations were estimated. For each population, mean (T_{mean}) and maximum (T_{max}) temperatures for each month were uniformly assigned to all days within that month. The daily temperature cycle was then reconstructed using a sinusoidal function to model intra-day variations. Instead of relying on maximum and minimum temperature for amplitude calculations, which could distort the daily mean, the amplitude was determined as $T_{max} - T_{mean}$. This approach ensured that simulated hourly temperatures oscillated around T_{mean} without introducing systematic bias. Hourly temperatures (T_{hour}) were computed using the equation:

(Eqn. 4) $T_{hour} = T_{mean} + (T_{max} - T_{mean}) \times \cos\frac{2\pi t}{24}$

where *t* is time in hours, and the cosine function captures the daily temperature variations.

From T_{hour}, the hourly heat increment (H_{inc}) was calculated using a non-linear thermal response function derived from the *GRT.Ex* model (Catara et al., 2016; Masin et al., 2017). This function, incorporating an exponential switch-off term, quantifies the effective thermal energy available to promote germination. It is defined as:

(Eqn. 5) $H_{inc} = \left\{ \begin{matrix} 0, \\ \frac{(T_{hour} - T_b)[1-\exp\{k(T_{hour}-T_c)\}}{1-\exp\{k(T_b-T_c)\}}, \\ 0, \end{matrix} \middle| \begin{matrix} T_{hour} < T_b, \\ (T_b \leq T_{hour} < T_c) \\ T_{hour} \geq T_c. \end{matrix} \right\}$

where *k* is the switch-off parameter controlling the negative effect of temperatures at supra-optimal levels. This equation indicates that temperatures below T_b and above T_c do not contribute to heat accumulation. For temperatures between T_b and T_c, the function quantifies the effective thermal energy available for germination while incorporating an exponential decay as temperatures approach T_c. The heat sum (Σ°Cd) is then given by the cumulative sum of all hourly heat increment, and the predicted germination timing to 50% corresponds to the first time point at which exceeds ϑ_T.

The heat sum accumulated by seeds in each population was calculated for each climate scenario. For non-stratified seeds, heat accumulation started after seed dispersal (i.e., the seed collection date). In contrast, for cold-stratified seeds, heat accumulation starts after winter, following a period of cold exposure during which seeds undergo stratification. Cold stratification requirements were considered met if the coldest month's mean temperature was ≤7°C (derived as 5°C, the mean temperature of the incubator used during cold stratification, plus an allowance of 2°C for the incubator's upper limit). When future projections indicated that conditions for cold stratification would not be met, the modelling was restricted to non-stratified seeds. Additionally, if T_b estimates for stratified and non-stratified seeds differed by <2 °C (which occurred for three *Alnus glutinosa* and all *Pinus sylvestris* collections), values were averaged for predictions, as this difference falls within the incubator's fluctuation range and was treated as experimental noise rather than a biologically meaningful effect. Both approaches mitigate the distortions of using monthly-averaged climate data, where slight T_b differences can translate into month-long shifts in the predicted days to reach 50 % germination.

Drivers of dormancy depth

Shifts in germination behaviour likely scale with dormancy depth, as populations with deeper primary dormancy should show greater delays or reductions in germination when environmental cues are unmet. As dormancy depth can be influenced both by local climate (Daws et al., 2004) and seed mass (Krzyszton et al., 2024), we used these factors as predictors of shift magnitude. First, we quantified dormancy depth as the stratification-induced shift in five thermal parameters (T_b, T_o, T_c, ϑ_T, and thermal window). For each seed lot, we calculated the difference between cold-stratified and non-stratified values (ΔT_b, ΔT_o, ΔT_c, $\Delta\vartheta_T$, Δthermal width), using these shifts as proxies for dormancy depth in subsequent analyses.

To characterise the climatic niche of the study populations while mitigating multicollinearity, we performed a Principal Component Analysis (PCA) on the 19 scaled bioclimatic variables and the mean temperature during dispersal (T_{disp}). T_{disp} was included to capture the specific environmental cues experienced by seeds at the moment of release and derived by averaging monthly mean temperatures over species-specific dispersal windows (*A. glutinosa*: November to January, *B. pubescens:* August to October, *P. sylvestris*: January to February). The first two principal components (PC1 and PC2) were retained to serve as orthogonal proxies for the local climatic niche in subsequent analyses.

To assess whether variation in dormancy depth is driven by climatic selection pressures or seed mass, we followed an Information-Theoretic approach using the *MuMIn* package (Bartoń, 2025). We fitted general linear models using two distinct predictor sets: 1) the proxies for climatic niche (n = 18) and 2) log-transformed seed mass (n = 15). For each response-predictor pair, we constructed linear models including predictor(s) and species identity. We generated all possible candidate model subsets using the 'dredge' function. Species identity was included as a fixed term in all candidate models to account for taxonomic variation in baseline trait values. Due to limited sample sizes, we restricted the global model structure to additive terms, excluding interaction terms to prevent overfitting and preserve statistical power for detecting main effects. Models were ranked based on the Akaike Information Criterion corrected for small sample sizes (AICc). To account for model selection uncertainty, we performed model averaging across a set of top-ranked models summing to a cumulative AICc weight of 0.95. Parameter estimates and 95% confidence intervals were calculated using full model averaging. The relative importance of predictors was assessed based on the magnitude of the averaged coefficients and whether their confidence intervals overlapped zero.

Fig. S1 Cumulative germination percentage over six weeks in response to temperature for the tested populations of *A. glutinosa* and *B. pubescens*, and *P. sylvestris*. COLD (cold-stratified) and NS (non-stratified).

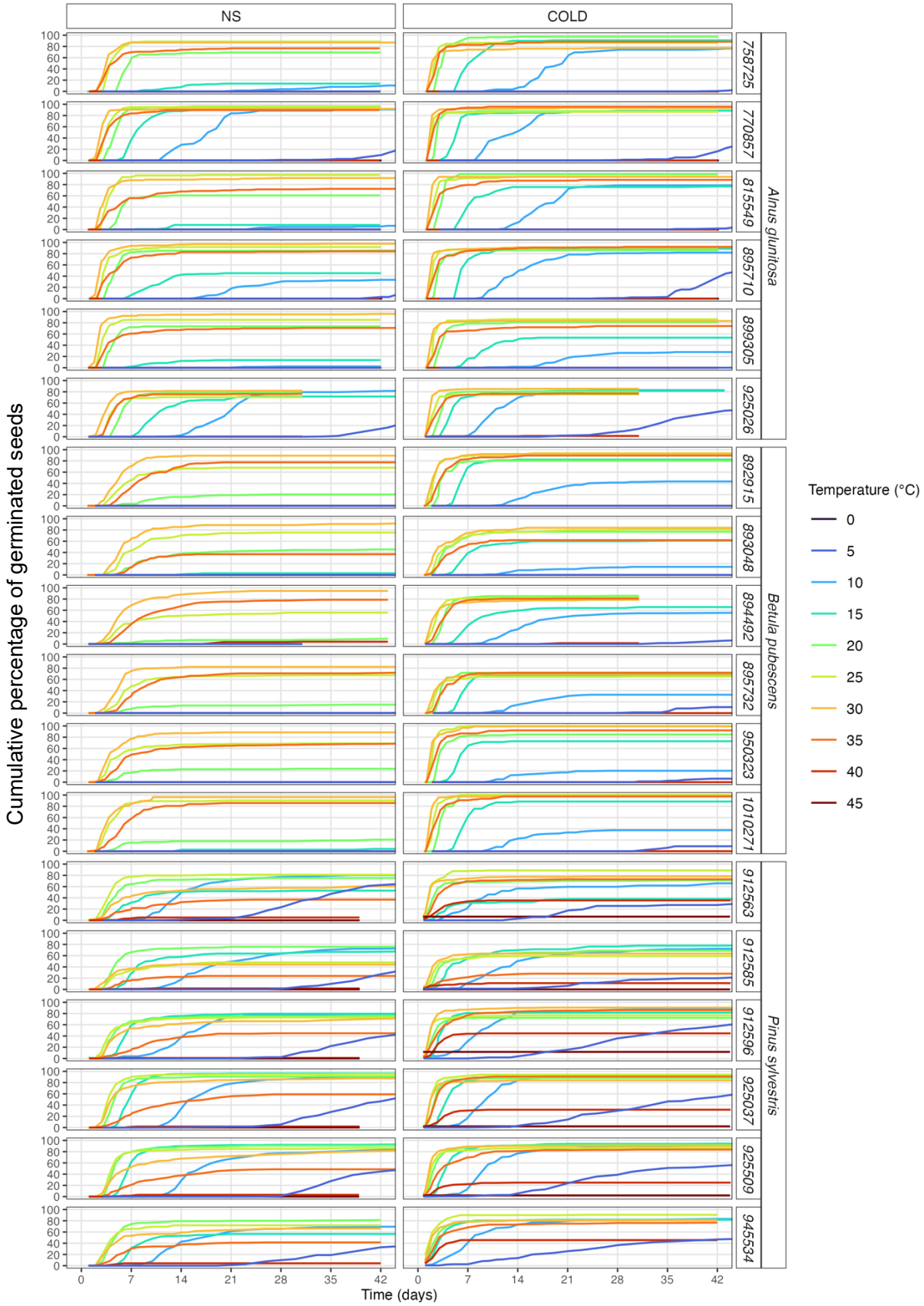
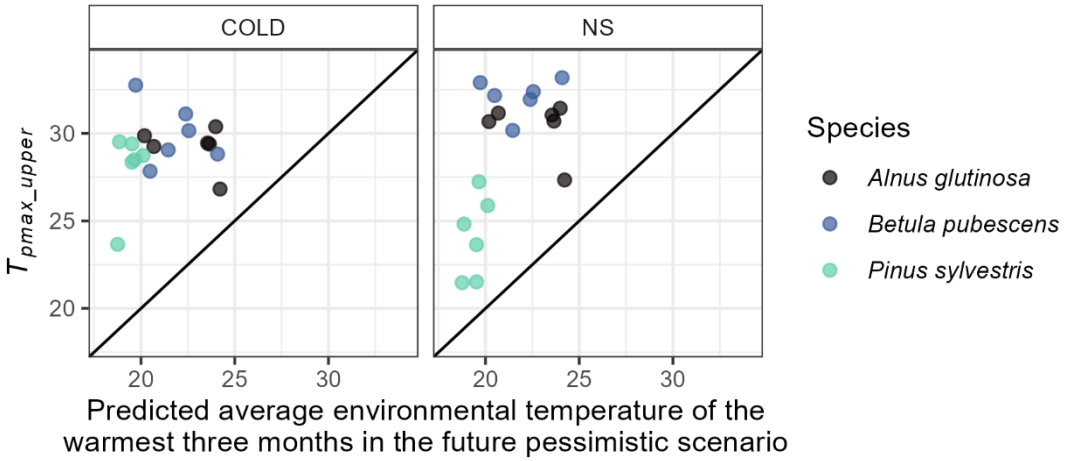
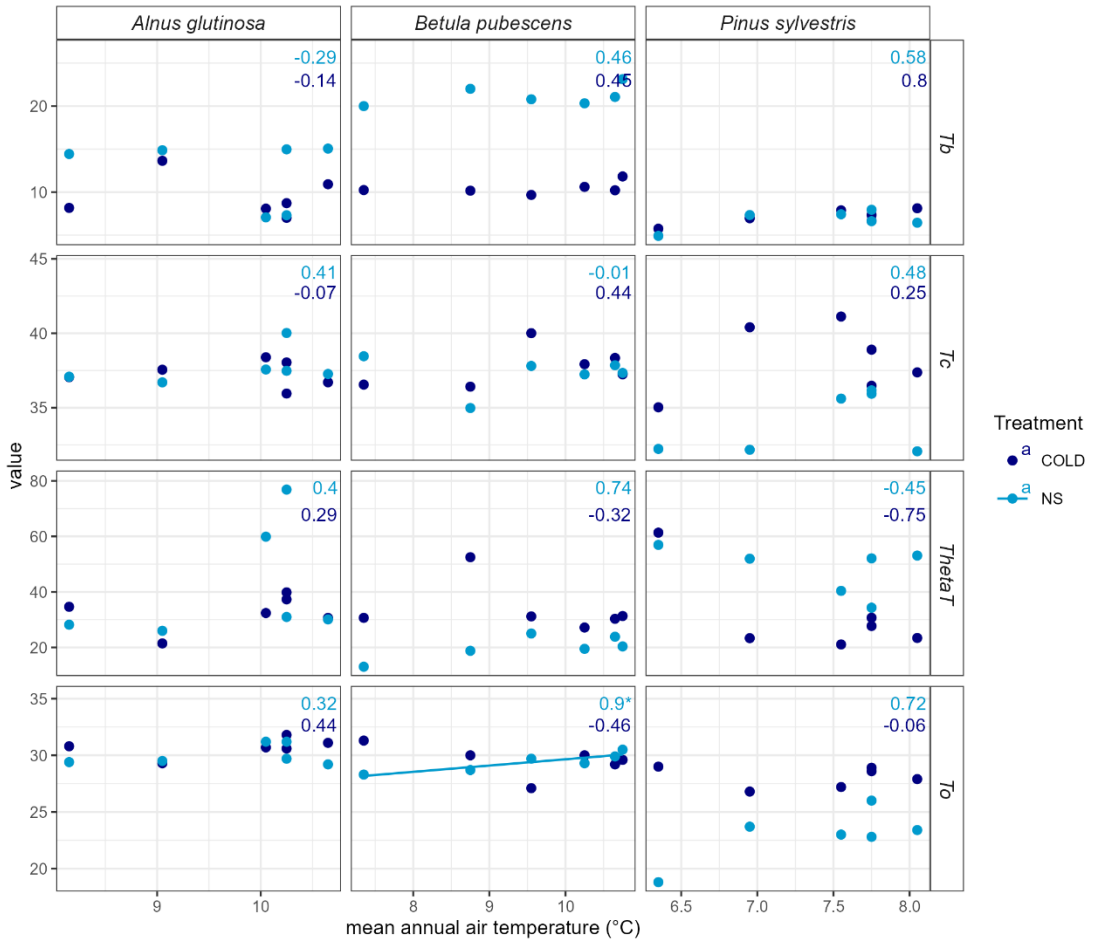


Fig. S2 Upper T_{pmax} value against the predicted average environmental temperature of the warmest three months under the future pessimistic scenario.



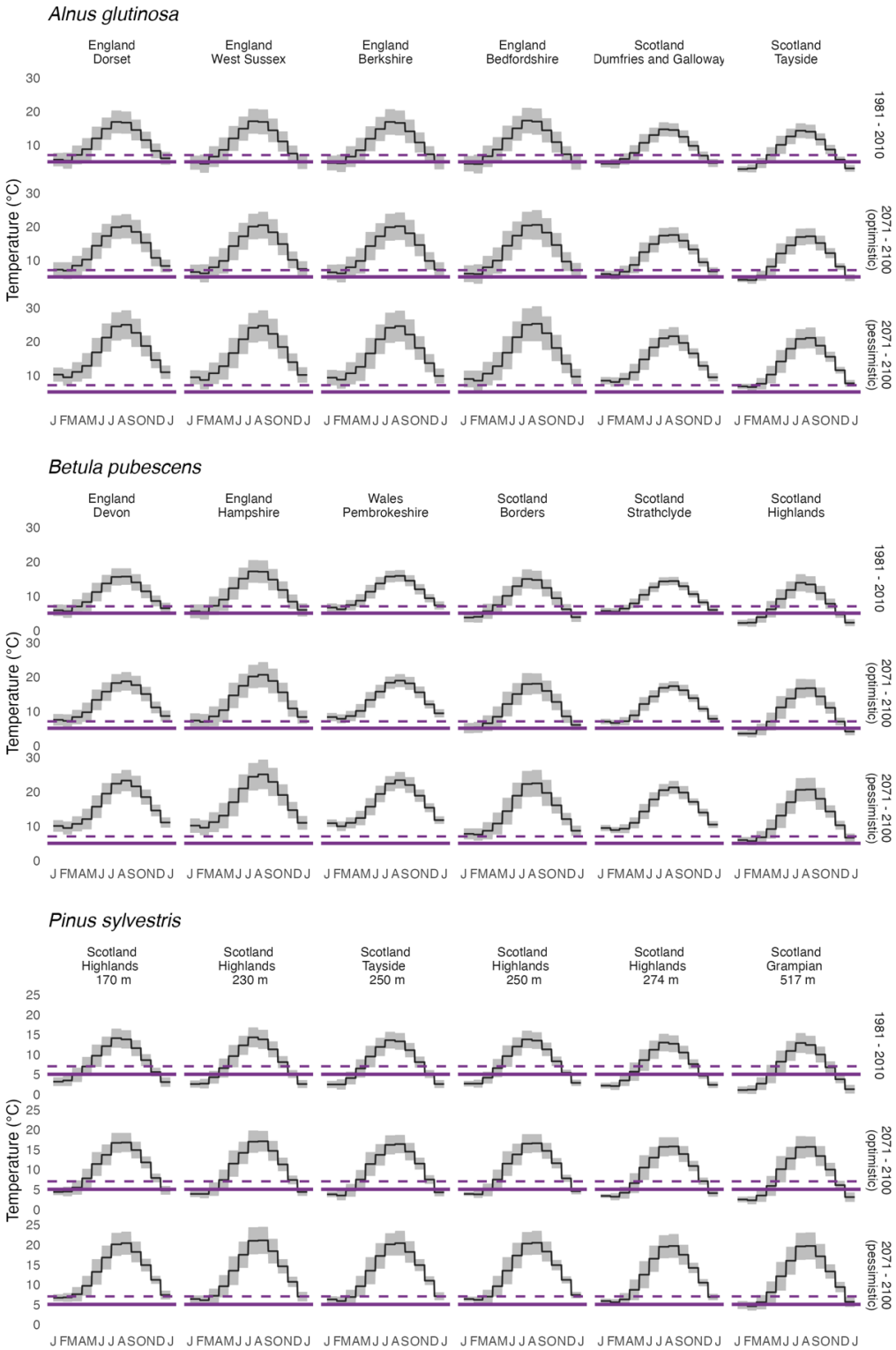
Each dot indicates one of the populations assessed. COLD (cold-stratified) and NS (non-stratified).

Fig. S3 Relationships between mean annual air temperature and four thermal parameters.



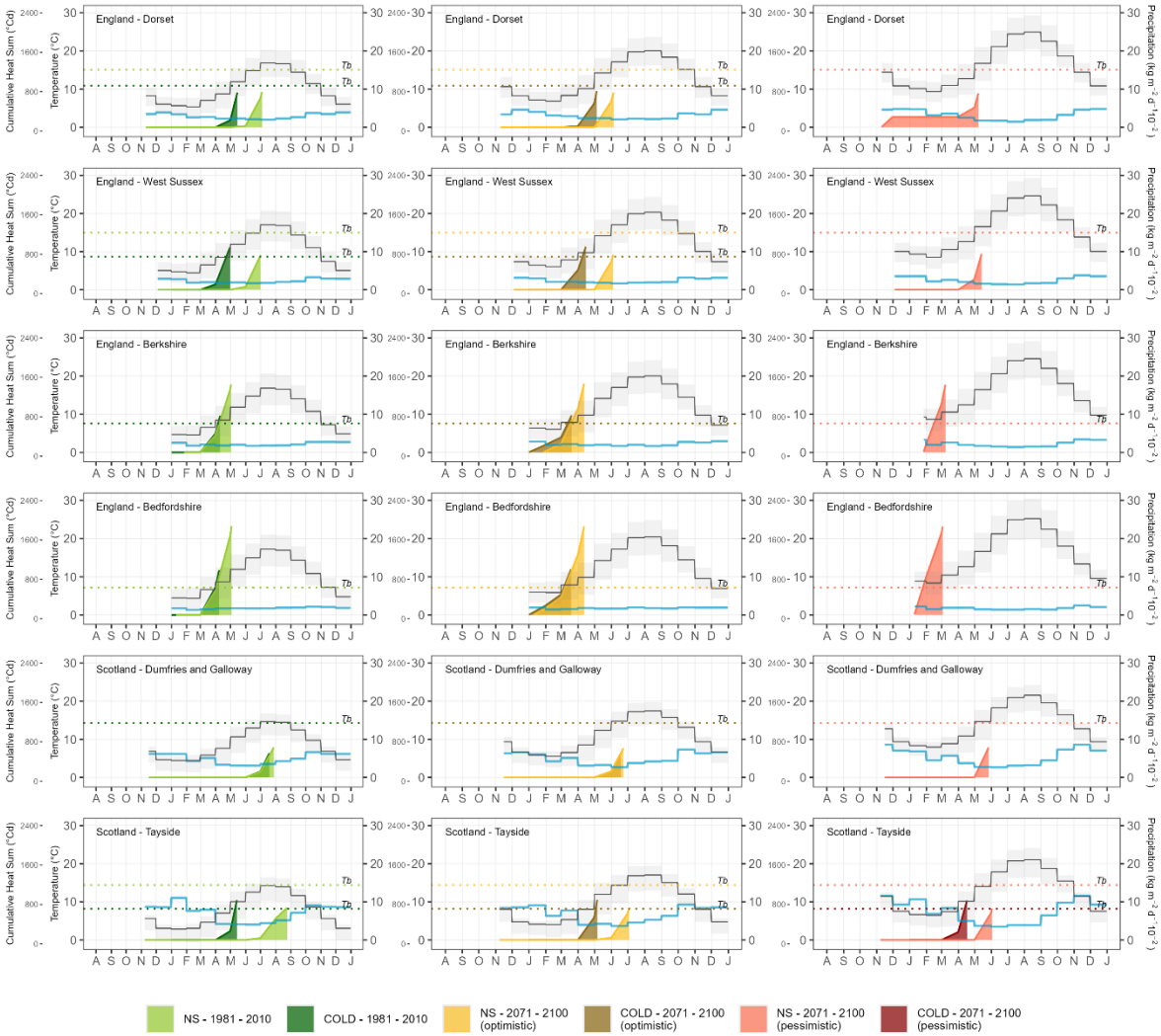
Relationships between mean annual air temperature and four thermal parameters (T_b, T_c, ϑ_T, T_o) for three species: *Alnus glutinosa, Betula pubescens, and Pinus sylvestris*. Points are individual populations under two treatments, COLD (cold-stratified) and NS (non-stratified). Numbers in the upper right of each panel are Pearson correlation coefficients for NS and COLD. An asterisk after a coefficient and the presence of a fitted line indicate a significant correlation (p<0.05).

Fig. S4 Average monthly temperatures for six populations of *Alnus glutinosa*, *Betula pubescens* and *Pinus sylvestris* under both current and projected UK climate scenarios.



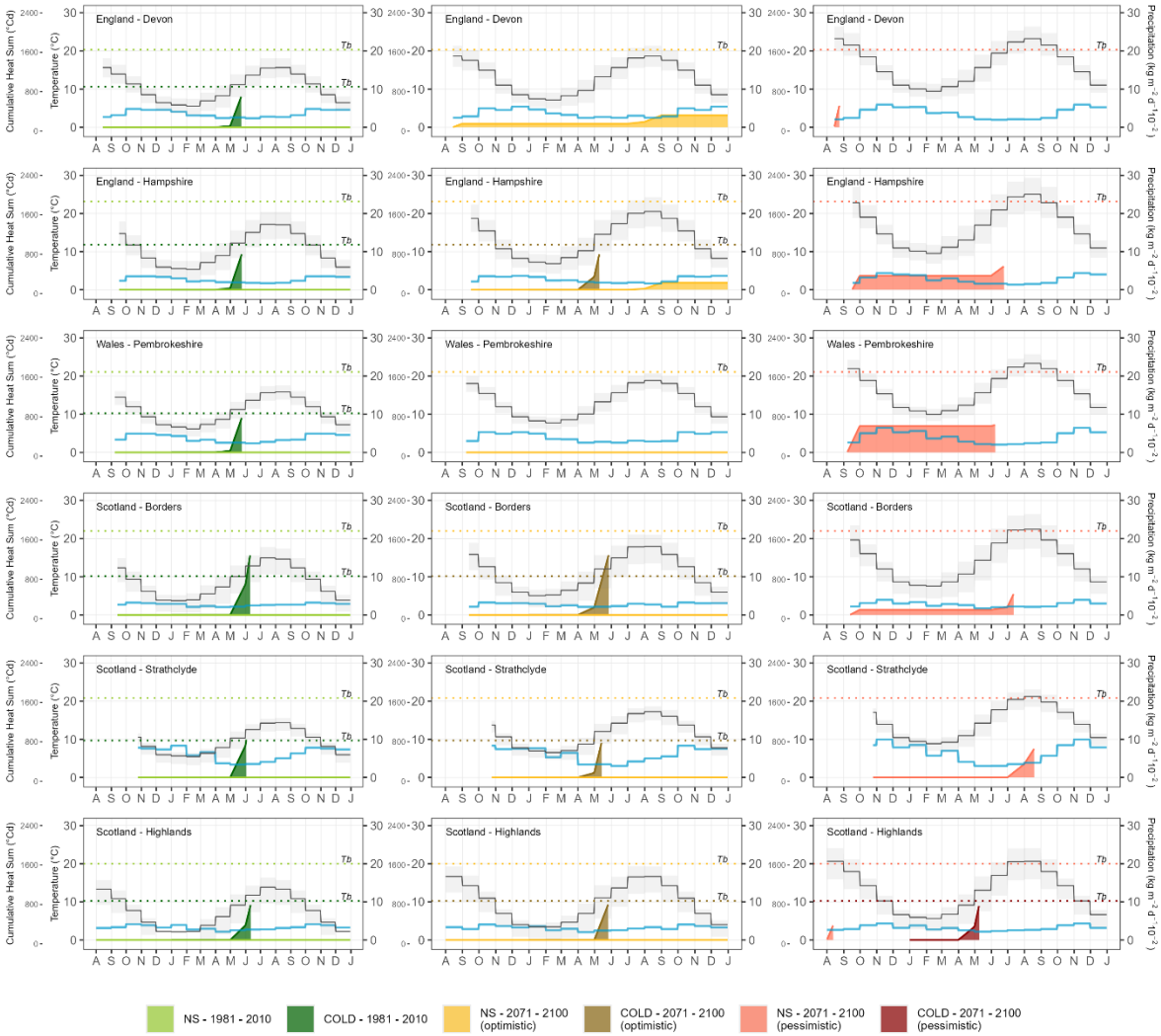
Monthly mean (solid black line), minimum, and maximum temperatures (shaded area). Cold stratification requirements were considered met if the average temperature during the coldest month for each population did not exceed 7°C (dashed purple line). This threshold reflects the incubator's average temperature of 5°C (solid line), with a variation of ±2°C.

Fig. S5 Environmental heat sum for non-stratified and cold-stratified seeds of *Alnus glutinosa*



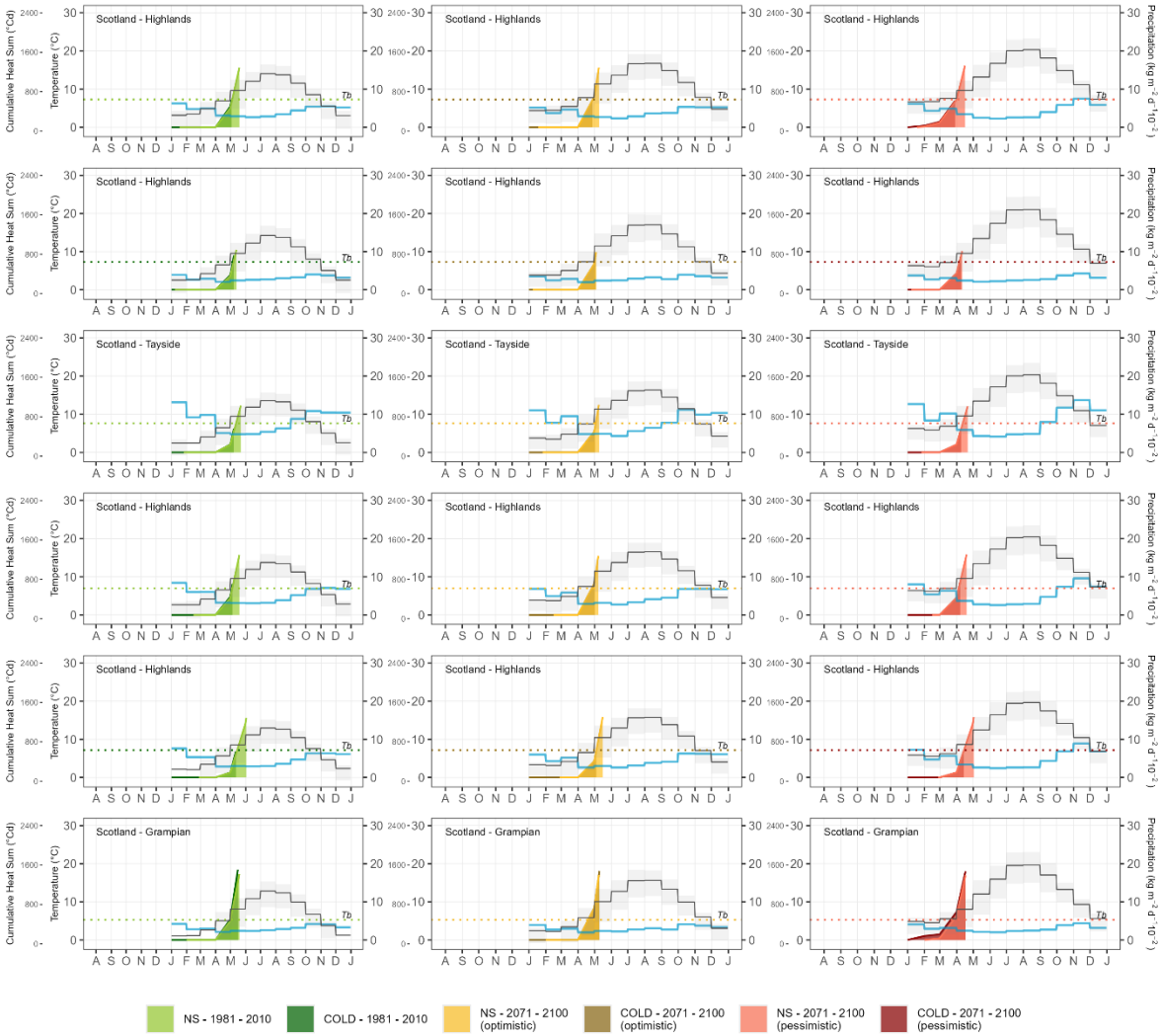
Environmental heat sum for non-stratified (NS) and cold-stratified (COLD) seeds of *Alnus glutinosa*, calculated for each seed lot and population using present-day climate data (1981–2010), and projected under optimistic (SSP2-2.6) and pessimistic (SSP5-8.5) climate scenarios. Cumulative heat sums were calculated up to the median thermal time required to reach 50% germination. Cold-stratified seeds were omitted from the plot when the requirement for cold stratification was not met. Dotted horizontal lines indicate the base temperatures for germination (T_b); note that T_b values for NS and COLD seeds may overlap. For each population, average monthly temperature (mean: black solid line; range: grey shaded area) and rainfall (blue solid line) are shown, starting from the month of seed collection. Seed lots are arranged along a latitudinal gradient.

Fig. S6 Environmental heat sum for non-stratified and cold-stratified seeds of *Betula pubescens*



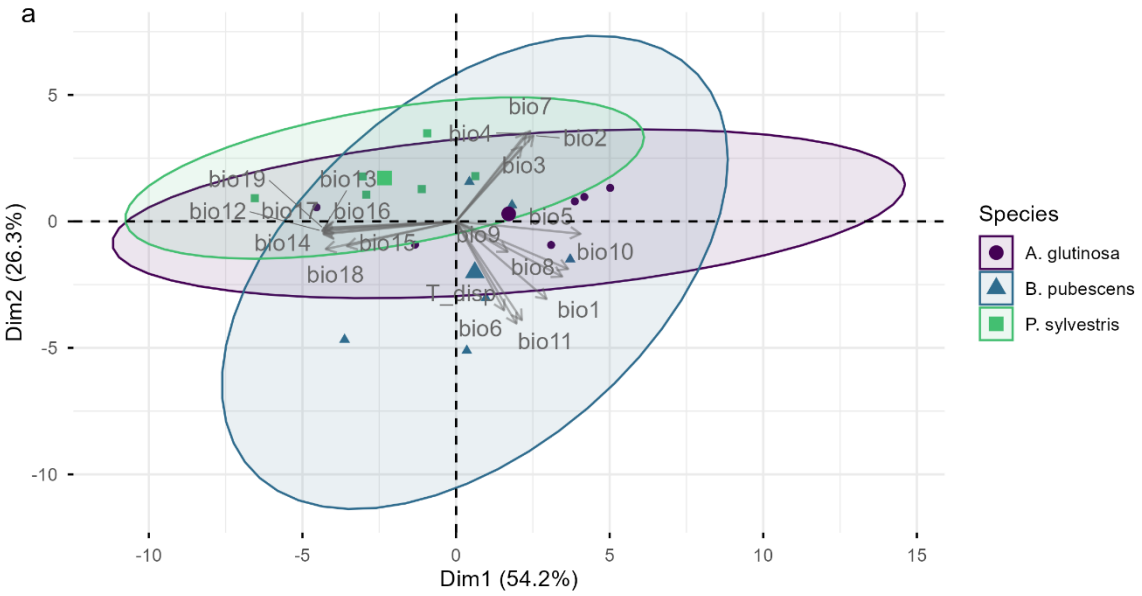
Environmental heat sum for non-stratified (NS) and cold-stratified (COLD) seeds of *Betula pubescens*, calculated for each seed lot and population using present-day climate data (1981–2010), and projected under optimistic (SSP2-2.6) and pessimistic (SSP5-8.5) climate scenarios. Cumulative heat sums were calculated up to the median thermal time required to reach 50% germination. Cold-stratified seeds were omitted from the plot when the requirement for cold stratification was not met. Dotted horizontal lines indicate the base temperatures for germination (T_b). For each population, average monthly temperature (mean: black solid line; range: grey shaded area) and rainfall (blue solid line) are shown, starting from the month of seed collection. Seed lots are arranged along a latitudinal gradient.

Fig. S7 Environmental heat sum for non-stratified and cold-stratified seeds of *Pinus sylvestris*



Environmental heat sum for non-stratified (NS) and cold-stratified (COLD) seeds of *Pinus sylvestris*, calculated for each seed lot and population using present-day climate data (1981–2010), and projected under optimistic (SSP2-2.6) and pessimistic (SSP5-8.5) climate scenarios. Cumulative heat sums were calculated up to the median thermal time required to reach 50% germination. Cold-stratified seeds were omitted from the plot when the requirement for cold stratification was not met. Dotted horizontal lines indicate the base temperatures for germination (T_b); note that T_b values for NS and COLD seeds may overlap. For each population, average monthly temperature (mean: black solid line; range: grey shaded area) and rainfall (blue solid line) are shown, starting from the month of seed collection. Seed lots are arranged along an altitudinal gradient.

Fig. S8 Principal Component Analysis (PCA) of climatic variables and dispersal-period temperature for seed collection sites.



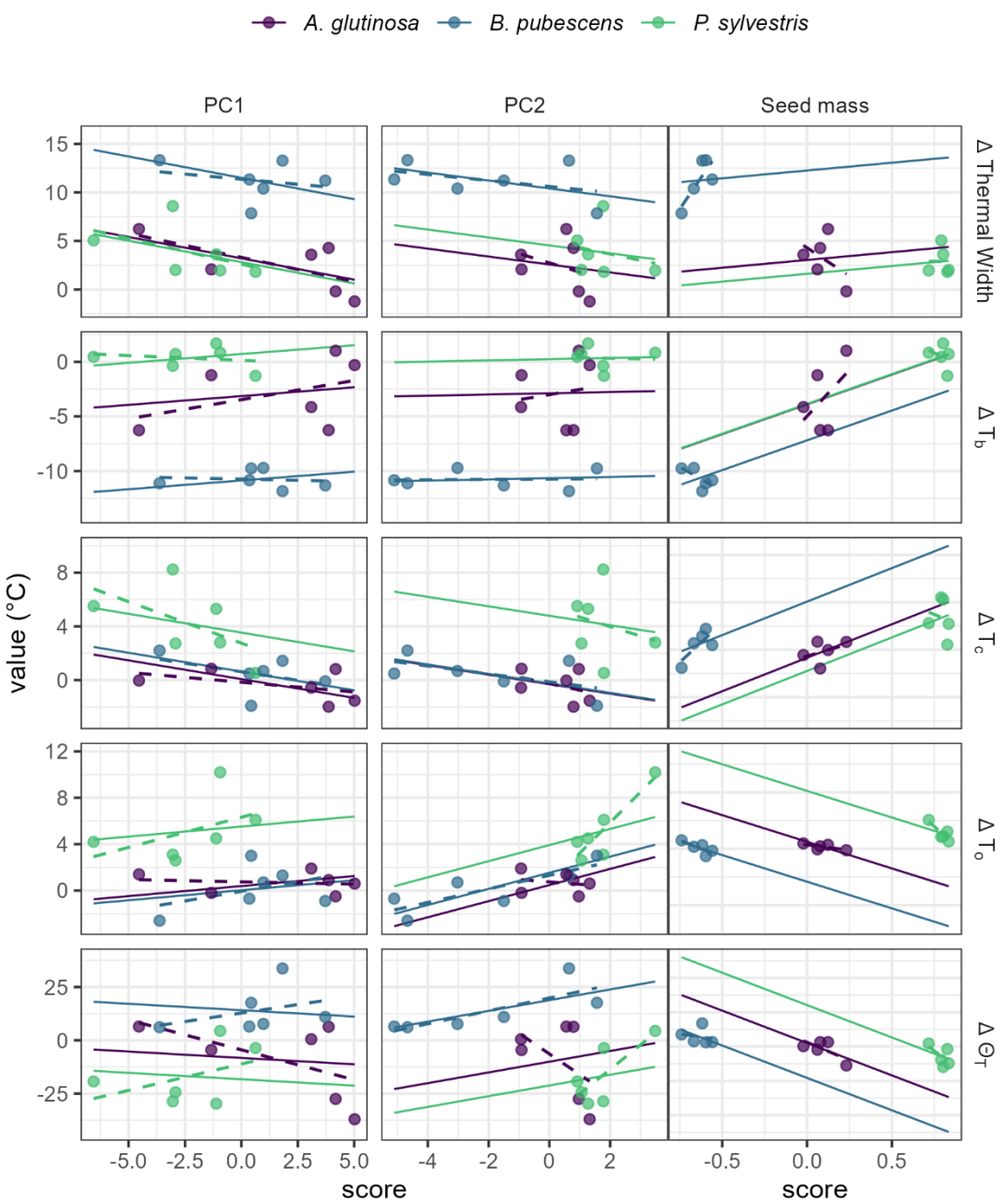
b

	PC1	PC2
mean daily mean air temperatures of the warmest quarter (°C) - bio10	0.79	-0.41
mean daily mean air temperatures of the wettest quarter (°C) - bio8	0.75	-0.47
mean annual air temperature (°C) - bio1	0.64	-0.67
mean daily mean air temperatures of the driest quarter (°C) - bio9	0.37	-0.26
mean daily maximum air temperature of the warmest month (°C) - bio5	0.88	-0.11
mean daily mean air temperatures of the coldest quarter (°C) - bio11	0.47	-0.85
mean daily minimum air temperature of the coldest month (°C) - bio6	0.43	-0.88
mean temperature during the seed dispersal months (°C) - T_disp	0.34	-0.76
annual range of air temperature (°C) - bio7	0.53	0.78
temperature seasonality (°C/100) - bio4	0.5	0.76
mean diurnal air temperature range (°C) - bio2	0.55	0.74
isothermality (°C) - bio3	0.47	0.64
mean monthly precipitation amount of the warmest quarter (kg m-2 month-1) - bio18	-0.92	-0.24
precipitation seasonality (kg m-2) - bio15	-0.77	-0.21
mean monthly precipitation amount of the driest quarter (kg m-2 month-1) - bio17	-0.95	-0.07
annual precipitation amount (kg m-2 year-1) - bio12	-0.95	-0.08
mean monthly precipitation amount of the coldest quarter (kg m-2 month-1) - bio19	-0.94	-0.06
precipitation amount of the wettest month (kg m-2 month-1) - bio13	-0.94	-0.05
mean monthly precipitation amount of the wettest quarter (kg m-2 month-1) - bio16	-0.94	-0.11
precipitation amount of the driest month (kg m-2 month-1) - bio14	-0.94	-0.1

Correlations: 1.0, 0.5, 0.0, -0.5, -1.0

(a) Distribution of populations in the main axes of the climatic niche multivariate space, PC1 vs PC2 (Dim1 vs Dim2). Points are individual populations, with species-specific 95% confidence ellipses. Arrows indicate loadings of bioclimatic variables and mean temperature during seed dispersal months. (b) Heatmap of correlation between variable loadings and PC1 and PC2. Variables include bioclimatic predictors (CHELSA dataset) and mean dispersal-period temperature.

Fig. S9 Shifts in germination thermal traits following cold stratification versus the proxies for local climatic niche and seed mass (mg, $\log_{10}$).



Shifts of base temperature (ΔT_b), optimum temperature (ΔT_o), ceiling temperature (ΔT_c), thermal time ($\Delta \vartheta_T$), and thermal width (Δ thermal width). Δ germination thermal traits for each seed lot were calculated by subtracting the cardinal temperature of cold-stratified seeds from that of non-stratified seeds. PC1 and PC2 from PCA on bioclimatic variables and mean dispersal-month temperature, served as proxies for local climatic niche (SI 11). Points represent observed data. Solid lines show additive model predictions (shared slope with species-specific intercepts), but no robust evidence of an effect was found. Dashed lines indicate species-specific fits for visual assessment only (model not formally tested). Parameter estimates derived from averaged models are reported in SI 14 and SI 15.

Table S1. Details of the seed lots used in the germination experiment.

Serial Number	Date Collected	Longitude	Latitude	Altitude (m)	Country	Region	Trees used for the experiments	Trees sampled *in situ*	Estimated population size	*MAAT* (°C)	P_{Ann} (kg m^{-2} year^{-1})	T_{maxW} (°C)	T_{minC} (°C)	T_{Range} (°C)	T_{disp} (°C)
Alnus glutinosa															
815549	10/11/2014	-2.9	50.8	80	England	Dorset	10	10	100	10.7	1001	20.3	2.55	18	6.7
758725	4/12/2013	-0.1	51.1	85	England	West Sussex	13	13	300	10.3	823	20.8	1.45	19	5.8
770857	27/01/2014	-1.5	51.4	95	England	Berkshire	5	12	40	10.1	768.6	20.7	1.45	19	5.6
925026*	10/01/2017	-0.5	52.0	50	England	Bedfordshire	11	11	32	10.3	637.5	21.1	1.45	20	5.5
899305	16/11/2016	-4.5	55.1	99	Scotland	Dumfries and Galloway	9	14	80	9.05	1704	16.8	1.35	15	5.3
895710	8/11/2016	-4.5	56.4	154	Scotland	Tayside	6	15	nd	8.15	2467	16.5	-0.5	17	3.9
Betula pubescens															
950323	15/08/2017	-4.3	50.9	150	England	Devon	8	10	50	10.3	1205	18.1	3.15	15	13.7
894492*	17/09/2016	-1.6	50.9	25	England	Hampshire	16	16	50	10.8	888	20.5	2.65	18	14.5
892915	8/09/2016	-4.8	51.8	30	Wales	Pembrokeshire	10	10	1000	10.7	1252	17.6	4.15	13	14.1
893048†	13/09/2016	-2.6	55.7	150	Scotland	Borders	11	15	2000	8.75	934.8	17.8	0.65	17	12.1
895732	25/10/2016	-5.2	55.9	60	Scotland	Strathclyde	9	21	100	9.55	2084	15.6	3.25	12	12.6
1010271	1/08/2018	-3.7	57.2	300	Scotland	Highlands	7	7	1000	7.35	1132	16.4	-1.3	18	10.7
Pinus sylvestris															
912563	18/01/2017	-4.6	58.0	170	Scotland	Highlands	7	9	nd	8.05	1537	16.5	-0.3	17	3.3
925509	8/01/2017	-3.7	57.2	230	Scotland	Highlands	12	19	1000	7.75	1103	16.8	-1	18	2.6
912596	26/01/2017	-4.7	56.4	250	Scotland	Tayside	15	17	nd	7.55	2965	15.7	-1	17	2.5
925037	15/02/2017	-4.8	57.2	250	Scotland	Highlands	8	15	30	7.75	1884	16	-0.8	17	2.7
945534*	26/02/2017	-4.6	57.2	274	Scotland	Highlands	13	16	nd	6.95	1714	15.2	-1.4	17	2.1
912585	1/02/2017	-3.5	57.0	517	Scotland	Grampian	9	18	nd	6.35	1115	15.4	-2.5	18	1.1

* Collections used during the first phase of the experiment. † Population identified as containing mixed *Betula* species based on ploidy analysis. *MAAT* (bio1) = mean annual air temperature (°C); P_{Ann} (bio12) = annual precipitation amount (kg m^{-2} year^{-1}); T_{maxW} (bio5) = mean daily maximum air temperature of the warmest month (°C); T_{minC} (bio6) = mean daily minimum air temperature of the coldest month (°C); T_{Range} (bio7) = annual range of air temperature (°C); T_{disp} = mean temperature during seed dispersal period (°C).

Table S2 Threshold temperatures and thermal requirements for seed lots from eighteen populations, with standard errors in parentheses. Abbreviations: T_b = base temperature (°C); T_c = ceiling temperature (°C); T_o = optimal temperature (°C); k = switch-off parameter (dimensionless); ϑ_T = thermal time to 50% germination according to the exponential model (°C·days); T_{pmax} = temperature at which maximum germination rate occurs (°C); T_{pmax} lower and T_{pmax} upper = 95% confidence interval limits for T_{pmax} (°C); max GR = maximum germination rate (d^{-1}). Treatments: COLD = cold-stratified seeds; NS = non-stratified seeds. * Collections used during the first phase of the experiment.

Species	Population	Serial Number	Treat	T_{pmax}	T_{pmax} lower	T_{pmax} upper	T_b	T_c	T_o	ϑ_T	k	max GR
Alnus glutinosa	England, Dorset	815549	NS	27.3	23.16	31.44	15.06 (1.4)	37.27 (0.64)	29.2	30.1 (8.6)	0.13 (0.081)	0.32
			COLD	22.71	15.04	30.38	10.91 (0.6)	36.71 (0.42)	31.1	30.63 (2.1)	0.387 (0.114)	0.58
	England, West Sussex	758725	NS	26.6	22.5	30.7	14.97 (0.17)	40.02 (0.1)	29.7	30.97 (1.3)	0.065 (0.008)	0.29
			COLD	22.22	15.05	29.39	8.71 (0.5)	38.04 (0.64)	30.6	37.32 (2.3)	0.251 (0.064)	0.5
	England, Berkshire	770857	NS	21.74	12.44	31.04	7.07 (0.48)	37.57 (0.52)	31.2	59.9 (2.5)	0.357 (0.08)	0.36
			COLD	21.5	13.54	29.45	8.08 (0.34)	38.39 (0.5)	30.7	32.39 (1.3)	0.247 (0.044)	0.59
	England, Bedfordshire	925026*	NS	21.85	16.36	27.34	7.31 (0.58)	37.48 (0.56)	31.2	76.87 (3.8)	0.359 (0.087)	0.28
			COLD	20.33	13.83	26.82	7.01 (0.56)	35.95 (0.25)	31.8	39.84 (1.6)	0.698 (0.18)	0.59
	Scotland, Dumfries and Galloway	899305	NS	26.94	22.71	31.16	14.87 (0.26)	36.7 (0.24)	29.5	25.99 (1.8)	0.181 (0.036)	0.42
			COLD	24.51	19.76	29.25	13.64 (0.39)	37.55 (0.43)	29.3	21.44 (2)	0.144 (0.039)	0.52
	Scotland, Tayside	895710	NS	25.17	19.67	30.67	14.43 (0.26)	37.08 (0.28)	29.4	28.18 (2)	0.157 (0.033)	0.38
			COLD	21.92	13.97	29.87	8.17 (0.4)	37.05 (0.37)	30.8	34.64 (1.4)	0.353 (0.073)	0.58
Betula pubescens	England, Devon	950323	NS	29	26.05	31.94	20.33 (16)	37.24 (5.3)	29.3	19.49 (86)	0.03 (0.912)	0.24
			COLD	24.66	18.22	31.11	10.61 (0.73)	37.92 (0.57)	30	27.17 (2.8)	0.2 (0.057)	0.57
	England, Hampshire	894492*	NS	30.36	27.54	33.18	23.14 (5.8)	37.34 (8.6)	30.5	20.36 (75)	0.023 (1.186)	0.19
			COLD	23.44	18.05	28.82	11.82 (0.97)	37.24 (0.65)	29.6	31.32 (5.1)	0.197 (0.084)	0.44
	Wales, Pembrokeshire	892915	NS	29.33	26.28	32.39	21.06 (9.9)	37.85 (6.2)	29.9	23.85 (77)	0.024 (0.747)	0.19
			COLD	23.85	17.55	30.16	10.22 (0.87)	38.34 (0.64)	29.2	30.32 (4)	0.146 (0.047)	0.47

Species	Population	Serial Number	Treat	T_{pmax}	T_{pmax} lower	T_{pmax} upper	T_b	T_c	T_o	ϑ_T	k	max GR
	Scotland, Borders	893048	NS	27.27	24.36	30.17	22.01 (6.3)	34.98 (0.17)	28.7	18.78 (46)	0.018 (0.556)	0.18
			COLD	24.62	20.19	29.05	10.17 (2.2)	36.41 (0.78)	30	52.5 (13)	0.3 (0.199)	0.32
	Scotland, Strathclyde	895732	NS	29.43	26.71	32.16	20.8 (8.3)	37.81 (4.5)	29.7	25.02 (63)	0.023 (0.56)	0.19
			COLD	23.36	18.88	27.84	9.67 (0.69)	40.01 (0.48)	27.1	31.17 (4.7)	0.045 (0.023)	0.33
	Scotland, Highlands	1010271	NS	28.88	24.86	32.9	20 (0.25)	38.46 (1.1)	28.3	13.04 (2)	-0.041 (0.04)	0.29
			COLD	23.66	14.57	32.75	10.24 (0.42)	36.55 (0.3)	31.3	30.65 (1.3)	0.447 (0.099)	0.62
Pinus sylvestris	Highlands, 170m	912563	NS	13.86	6.2	21.52	6.44 (1.5)	32.07 (0.87)	23.4	53.1 (15)	0.14 (0.093)	0.23
			COLD	23.07	16.74	29.4	8.12 (0.57)	37.37 (0.4)	27.9	23.4 (2.2)	0.14 (0.033)	0.63
	Highlands, 230m	925509	NS	17.96	10.04	25.88	7.94 (0.63)	35.94 (0.4)	22.8	34.32 (5.1)	0.018 (0.02)	0.23
			COLD	19.76	10.78	28.73	6.66 (0.8)	36.47 (0.3)	28.9	30.69 (2.7)	0.247 (0.058)	0.61
	Tayside, 250m	912596	NS	17.38	11.11	23.64	7.43 (0.89)	35.61 (0.4)	23	40.38 (7.5)	0.032 (0.026)	0.22
			COLD	20.11	11.87	28.36	7.88 (0.8)	41.12 (0.42)	27.2	21.07 (2.9)	0.044 (0.018)	0.55
	Highlands, 250m	925037	NS	18.26	9.28	27.24	6.63 (0.6)	36.17 (0.19)	26	52.1 (4.6)	0.117 (0.02)	0.27
			COLD	18.45	8.41	28.49	7.34 (0.53)	38.9 (0.58)	28.6	27.72 (2.2)	0.127 (0.028)	0.57
	Highlands, 274m	945534*	NS	19.57	14.32	24.82	7.33 (1.5)	32.18 (0.99)	23.7	51.97 (15)	0.139 (0.095)	0.23
			COLD	21.95	14.39	29.52	6.96 (0.98)	40.4 (0.34)	26.8	23.33 (3.5)	0.051 (0.019)	0.52
	Grampian, 517m	912585	NS	16.62	11.76	21.47	4.91 (1.4)	32.23 (2.2)	18.8	56.94 (20)	0.005 (0.056)	0.12
			COLD	18.56	13.46	23.66	5.74 (2.6)	35.03 (0.23)	29	61.35 (12)	0.378 (0.148)	0.34

Table S3 Model selection results for variation in dormancy depth in response to local climatic niche controlling for species effects.

Response	Model	DF	Log-likelihood	AICc	Δ AICc	AICc weight	Top-ranked
Δ Estimate T_b	baseline	4	-36.073	83.2	0	0.709	Y
	PC1	5	-35.527	86.1	2.8	0.172	Y
	PC2	5	-36.05	87.1	3.9	0.102	Y
	PC1 + PC2	6	-35.512	90.7	7.4	0.017	
Δ Estimate T_o	PC2	5	-32.512	80	0	0.804	Y
	baseline	4	-36.641	84.4	4.3	0.092	Y
	PC1 + PC2	6	-32.491	84.6	4.6	0.081	Y
	PC1	5	-36.044	87.1	7.1	0.023	
Δ Estimate T_c	baseline	4	-35.368	81.8	0	0.402	Y
	PC1	5	-33.452	81.9	0.1	0.384	Y
	PC2	5	-34.334	83.7	1.9	0.159	Y
	PC1 + PC2	6	-33.064	85.8	4	0.056	Y
Δ Estimate ϑ_T	baseline	4	-72.375	155.8	0	0.645	Y
	PC2	5	-71.52	158	2.2	0.213	Y
	PC1	5	-72.244	159.5	3.7	0.103	Y
	PC1 + PC2	6	-70.907	161.5	5.6	0.039	
Δ Estimate thermal width	PC1	5	-37.735	90.5	0	0.598	Y
	baseline	4	-40.543	92.2	1.7	0.257	Y
	PC2	5	-39.774	94.5	4.1	0.078	Y
	PC1 + PC2	6	-37.591	94.8	4.3	0.068	Y

Note: PC1 and PC2 from PCA on bioclimatic variables and mean dispersal-month temperature, served as proxies for local climatic niche. The delta (Δ) of germination thermal traits, considered as proxies for dormancy depth, corresponds to the difference between non-stratified and cold-stratified seeds for each seed lot (n = 6 seed lots for each of the three species). Traits: base temperature (T_b), optimum temperature (T_o), ceiling temperature (T_c), thermal time (ϑ_T), and thermal width. For each response-predictor pair, we constructed linear models including predictors for climatic niche and species identity (response ~ sp + PC1 + PC2). All possible candidate model subsets that included species identity (to account for taxonomic variation in baseline trait values) were ranked based on the Akaike Information Criterion corrected for small sample sizes (AICc). To account for model selection uncertainty, we performed model averaging across a set of top-ranked models summing to a cumulative AICc weight of 0.95. Parameter estimates and 95% confidence intervals were calculated using full model averaging. Each row reports the model structure, degrees of freedom (DF), log-likelihood, ΔAICc (difference from the best model), and AICc weight (relative model likelihood).

Table S4 Model selection results for variation in dormancy depth in response to seed mass (mg, $\log_{10}$) controlling for species (sp) effects.

Response	Model	DF	Log-likelihood	AICc	Δ AICc	AICc weight	Top-ranked
Δ Estimate T_b	baseline	4	-30.188	72.4	0	0.885	Y
	log(seed mass)	5	-29.897	76.5	4.1	0.115	Y
Δ Estimate T_o	log(seed mass)	5	-27.25	71.2	0	0.822	Y
	baseline	4	-31.113	74.2	3.1	0.178	Y
Δ Estimate T_c	baseline	4	-26.994	66	0	0.827	Y
	log(seed mass)	5	-26.226	69.1	3.1	0.173	Y
Δ Estimate ϑ_T	baseline	4	-58.615	129	0	0.529	Y
	log(seed mass)	5	-56.396	130	0.2	0.471	Y
Δ Estimate thermal width	baseline	4	-30.599	73.2	0	0.91	Y
	log(seed mass)	5	-30.576	77.8	4.6	0.09	Y

Note: The delta (Δ) of germination thermal traits, considered as proxies for dormancy depth, corresponds to the difference between non-stratified and cold-stratified seeds for each seed lot (n = 5 seed lots for each of the three species). Traits: base temperature (T_b), optimum temperature (T_o), ceiling temperature (T_c), thermal time (ϑ_T), and thermal width. For each response-predictor pair, we constructed linear models including log seed mass and species identity (response ~ sp + log seed mass). The other candidate model included only species identity (to account for taxonomic variation in baseline trait values). Both models were ranked based on the Akaike Information Criterion corrected for small sample sizes (AICc). To account for model selection uncertainty, we performed model averaging across a set of top-ranked models summing to a cumulative AICc weight of 0.95. Parameter estimates and 95% confidence intervals were calculated using full model averaging. Each row reports the model structure, degrees of freedom (DF), log-likelihood, ΔAICc (difference from the best model), and AICc weight (relative model likelihood).

Table S5 Slope estimates for the effects of bioclimatic principal components PC1 and PC2 on dormancy depth derived from the averaged models.

Response	Predictor	Estimate	95% CI lower	95% CI upper	SE	Adjusted SE	z-value	p-value
Δ Estimate T_b	**(Intercept)**	**-2.91**	**-4.67**	**-1.16**	**0.82**	**0.9**	**3.26**	**0.001**
	B. pubescens	**-7.86**	**-10.35**	**-5.37**	**1.17**	**1.27**	**6.19**	**<0.001**
	P. sylvestris	**3.31**	**0.73**	**5.88**	**1.21**	**1.32**	**2.52**	**0.012**
	PC1	0.03	-0.17	0.22	0.09	0.1	0.28	0.778
	PC2	0.01	-0.2	0.21	0.1	0.11	0.05	0.957
Δ Estimate T_o	(Intercept)	0.49	-1.03	2.01	0.71	0.78	0.63	0.526
	B. pubescens	0.91	-1.68	3.49	1.22	1.32	0.69	0.491
	P. sylvestris	**3.55**	**1.2**	**5.91**	**1.1**	**1.2**	**2.96**	**0.003**
	PC1	0	-0.1	0.1	0.05	0.05	0.05	0.964
	PC2	0.63	-0.01	1.27	0.31	0.33	1.92	0.055
Δ Estimate T_c	(Intercept)	-0.18	-1.88	1.52	0.8	0.87	0.21	0.838
	B. pubescens	0.58	-1.84	3.01	1.14	1.24	0.47	0.637
	P. sylvestris	**4.2**	**1.44**	**6.96**	**1.31**	**1.41**	**2.98**	**0.003**
	PC1	-0.12	-0.47	0.23	0.17	0.18	0.68	0.496
	PC2	-0.07	-0.44	0.31	0.18	0.19	0.35	0.724
Δ Estimate ϑ_T	(Intercept)	-9.35	-22.38	3.69	6.11	6.65	1.41	0.16
	B. pubescens	**24.27**	**4.75**	**43.8**	**9.19**	**9.96**	**2.44**	**0.015**
	P. sylvestris	-8.63	-27.72	10.46	8.96	9.74	0.89	0.376
	PC1	-0.06	-1.06	0.93	0.47	0.51	0.13	0.899
	PC2	0.56	-2.41	3.53	1.45	1.51	0.37	0.713
Δ Estimate thermal width	**(Intercept)**	**2.97**	**0.74**	**5.2**	**1.05**	**1.14**	**2.61**	**0.009**
	B. pubescens	**8.36**	**5.32**	**11.39**	**1.42**	**1.55**	**5.39**	**<0.001**
	P. sylvestris	0.27	-3.42	3.97	1.76	1.89	0.15	0.884
	PC1	-0.29	-0.82	0.24	0.26	0.27	1.07	0.283
	PC2	-0.04	-0.41	0.33	0.18	0.19	0.23	0.821

Note: The delta (Δ) of germination thermal traits corresponds to the difference between non-stratified and cold-stratified seeds for each seed lot. Traits: base temperature (T_b), optimum temperature (T_o), ceiling temperature (T_c), thermal time (ϑ_T), and thermal width. Model averaging was carried out across the top-ranked models summing to a cumulative AICc weight of 0.95. Slope estimate (Estimate), confidence interval (CI); standard error (SE), test statistic (z-value). Estimates represent the 'full average' (shrinkage estimates), where coefficients are set to zero in component models where the predictor is absent. The intercept represents the estimated trait value for the reference species (*Alnus glutinosa*) under average climatic conditions. Coefficients for other species represent the estimated difference relative to *A. glutinosa*. Confidence intervals that exclude zero (bolded) are evidence of an effect; marginal evidence of an effect (underlined).

Table S6 Slope estimates for the effects of seed mass (mg, $\log_{10}$) on dormancy depth derived from the averaged models.

Response	Predictor	Estimate	95% CI lower	95% CI upper	SE	Adjusted SE	z-value	p-value
Δ Estimate T_b	**(Intercept)**	**-3.44**	**-5.53**	**-1.35**	**0.96**	**1.07**	**3.22**	**0.001**
	B. pubescens	**-6.82**	**-12.69**	**-0.96**	**2.73**	**2.99**	**2.28**	**0.023**
	P. sylvestris	3.42	-2.26	9.09	2.64	2.9	1.18	0.238
	log(seed mass)	0.63	-6.42	7.67	3.3	3.6	0.17	0.862
Δ Estimate T_o	(Intercept)	2.17	-0.42	4.75	1.22	1.32	1.64	0.101
	B. pubescens	-11.7	-26.39	2.99	7.09	7.49	1.56	0.118
	P. sylvestris	**15.69**	**1.59**	**29.78**	**6.81**	**7.19**	**2.18**	**0.029**
	log(seed mass)	-15.52	-35.32	4.28	9.58	10.1	1.54	0.125
Δ Estimate T_c	(Intercept)	-0.29	-2.06	1.47	0.81	0.9	0.33	0.743
	B. pubescens	1.65	-4.55	7.85	2.96	3.16	0.52	0.602
	P. sylvestris	2.7	-3.28	8.68	2.85	3.05	0.89	0.376
	log(seed mass)	1.22	-6.69	9.13	3.79	4.03	0.3	0.762
Δ Estimate ϑ_T	(Intercept)	0.45	-16.34	17.24	7.92	8.57	0.05	0.958
	B. pubescens	-14.39	-102.22	73.43	42.87	44.81	0.32	0.748
	P. sylvestris	20.33	-63.99	104.65	41.15	43.02	0.47	0.637
	log(seed mass)	-44.42	-162.11	73.27	57.6	60.05	0.74	0.459
Δ Estimate thermal width	**(Intercept)**	**3.18**	**1.07**	**5.29**	**0.97**	**1.08**	**2.96**	**0.003**
	B. pubescens	**8.15**	**3.02**	**13.27**	**2.34**	**2.61**	**3.12**	**0.002**
	P. sylvestris	-0.4	-5.38	4.58	2.27	2.54	0.16	0.874
	log(seed mass)	0.14	-5.65	5.94	2.64	2.96	0.05	0.961

Note: The delta (Δ) of germination thermal traits corresponds to the difference between non-stratified and cold-stratified seeds for each seed lot. Traits: base temperature (T_b), optimum temperature (T_o), ceiling temperature (T_c), thermal time (ϑ_T), and thermal width. Model averaging was carried out across the top-ranked models summing to a cumulative AICc weight of 0.95. Slope estimate (Estimate), confidence interval (CI); standard error (SE), test statistic (z-value). Estimates represent the 'full average' (shrinkage estimates), where coefficients are set to zero in component models where the predictor is absent. The intercept represents the estimated trait value for the reference species (*Alnus glutinosa*) under average climatic conditions. Coefficients for other species represent the estimated difference relative to *A. glutinosa*. Confidence intervals that exclude zero (bolded) are evidence of an effect; marginal evidence of an effect (underlined).

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